

A top-down view of a grey desk with various items: a silver stethoscope, a laptop displaying a medical website, a pair of glasses, a black smartphone, a white coffee cup, and a spiral-bound notebook with a black pen. A large, semi-transparent white pill shape is centered on the desk.

Pharm*Achieve*

DRUG INTERACTIONS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ARB Angiotensin Receptor Blocker

ASA Aspirin

CYP Cytochrome

FQ Fluoroquinolone

GI Gastrointestinal

H2RA Histamine Type-2 Receptor Antagonist

Mino Minocycline

NNRTI Non-Nucleoside Reverse Transcriptase Inhibitor

NRTI Nucleoside (or Nucleotide) Reverse Transcriptase Inhibitor

PCN Penicillin

PGP P-Glycoprotein

PPI Proton Pump Inhibitor

TCA Tricyclic Antidepressant

DEFINITIONS

Terminology	Definition
Drug interactions	Quality, state, or action of ≥ 2 drugs acting on each other
Pharmacokinetics	Describes what the body does to a drug as the drug moves into, throughout, and out of the body Described as a drug's: absorption, distribution, metabolism, excretion
Pharmacodynamics	Describes how the drug affects the body

ABSORPTION: COMMON MECHANISMS

Chelation

- Mg^{2+} , Ca^{2+} , Fe^{3+} bind to drug and prevent GI absorption
- Examples: antibiotics (fluoroquinolones, tetracyclines), levothyroxine

pH dependent

- ↓absorption due to changes in gastric pH
- Examples: acid-reducers (H_2 RAs, PPIs), itraconazole, iron

Binding

- Similar to chelation but with non-cationic drugs
- Example: cholestyramine or charcoal use with multiple drugs

Efflux

- Mediated by p-glycoprotein thus altering intestinal absorption
- Examples: digoxin, quinidine

Intra-intestinal metabolism

- Phase I oxidation via CYP enzymes may affect amount of drug available for absorption
- Examples: grapefruit juice with 3A4 substrates

DISTRIBUTION



Note, distribution is less likely to be clinically significant for most drugs except for those with narrow therapeutic indexes

Protein binding

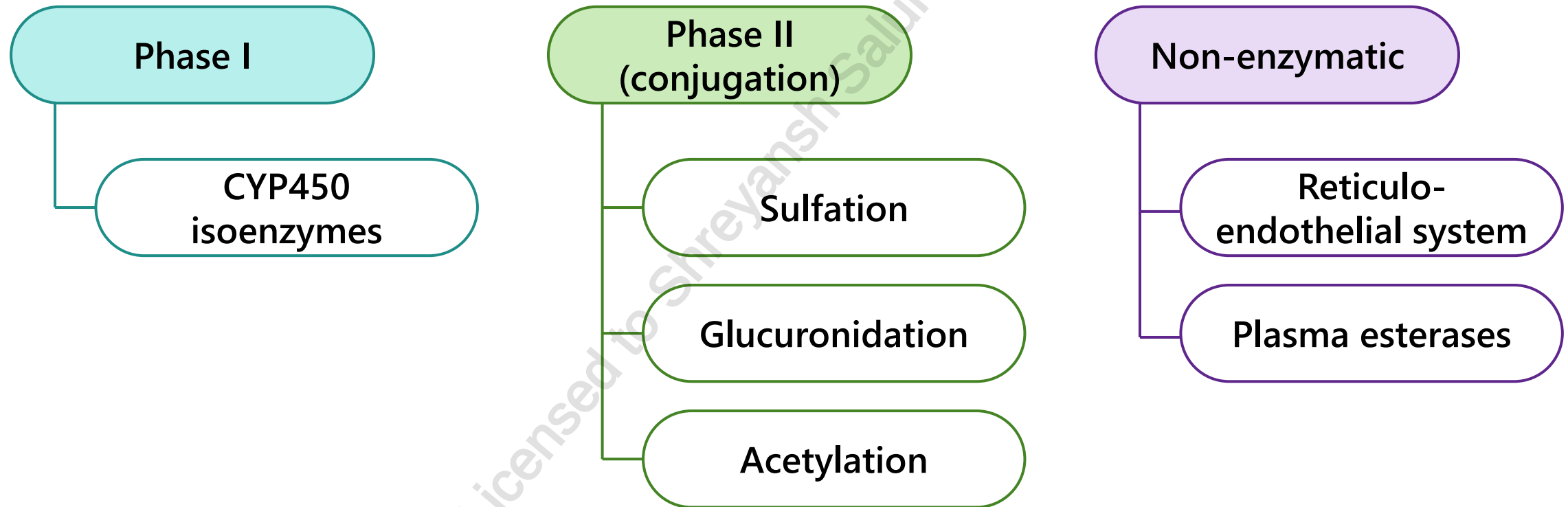
- Displacement reactions due to 2 highly protein-bound drugs may increase the free fraction of either drug
- Valproic acid and phenytoin
- Warfarin and ASA

PGP interactions

- Reduce cellular uptake of some drugs into CNS
- Examples: cancer chemotherapy (e.g. taxanes, vinca alkaloids), verapamil, loperamide, CD4 cell entry of protease inhibitors

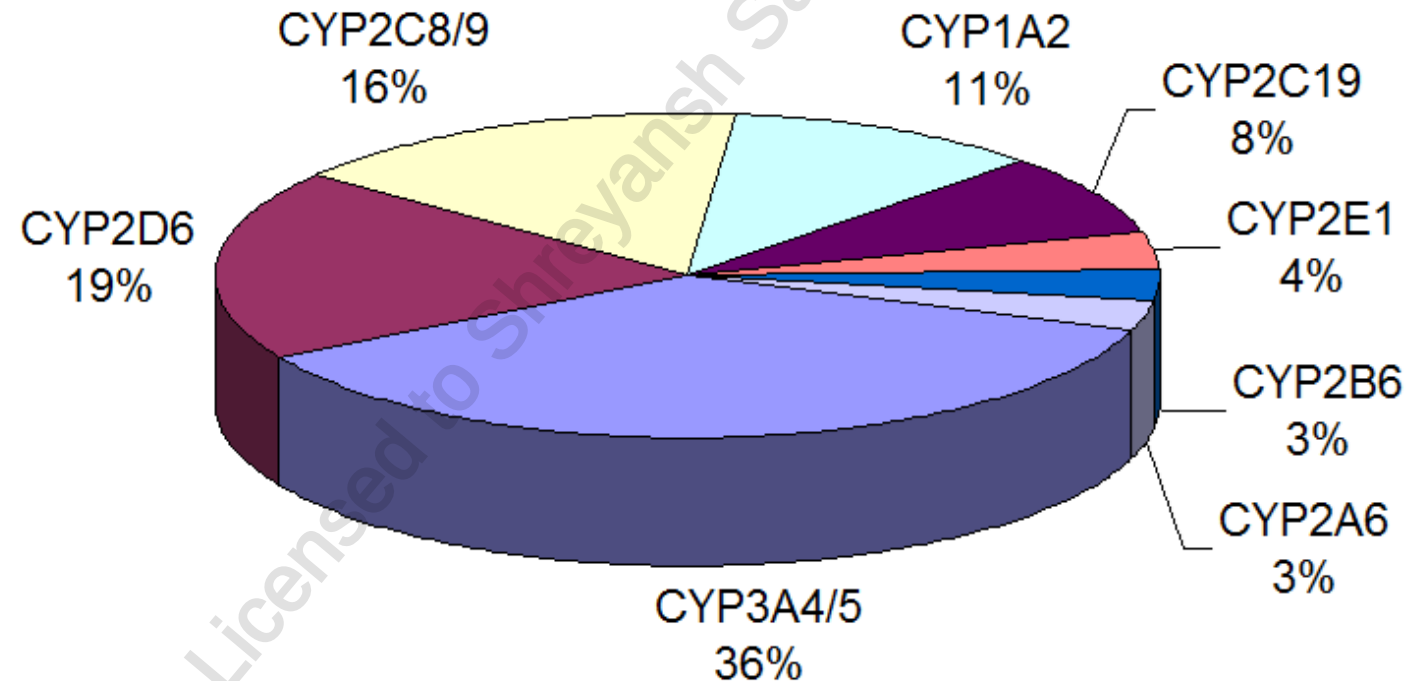
METABOLISM/BIOTRANSFORMATION

Most common reason for drug interactions; metabolism is primarily mediated by phase I and II enzymes while few drugs undergo non-enzymatic processes



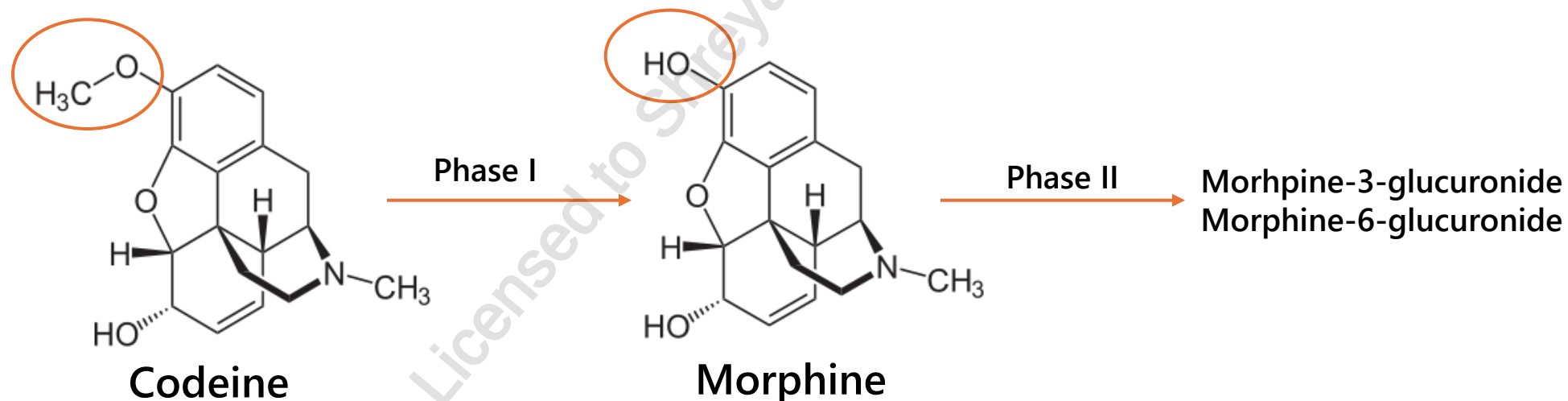
BIOTRANSFORMATION: PHASE I

- Polar group is introduced or unmasked, making the drug hydrophilic and easier to excrete
- Enzymes involved: cytochromes, oxidases, esterases and hydrolases
- Cytochromes are the most prevalent and most varied:



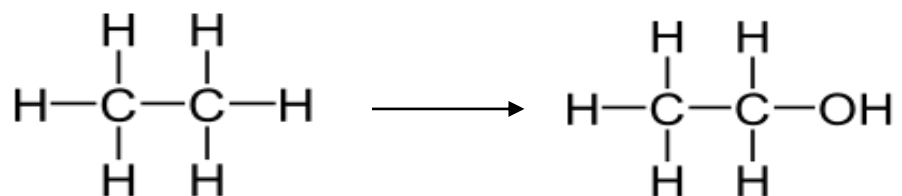
BIOTRANSFORMATION: PHASE II

- Drugs combine with substrates to undergo conjugation reactions, decreasing lipophilicity
- Most Phase-II reactions result in drug inactivation, where some products may be more active than parent compounds
 - Example: morphine metabolism through glucuronidation (*image*)
- Enzymes involved: different transferases for different substrates
 - Glutathione-s-transferases are most important

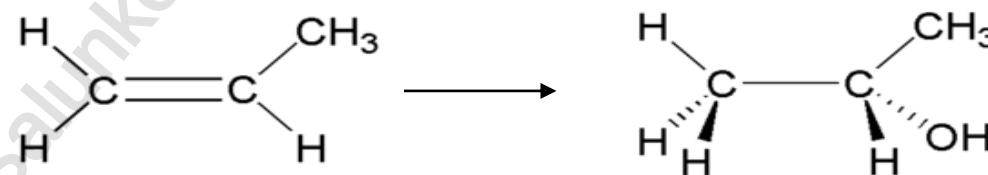


4 MAIN BIOTRANSFORMATION REACTIONS

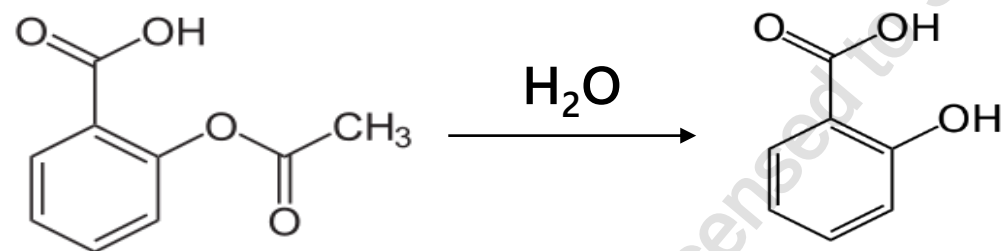
Oxidation



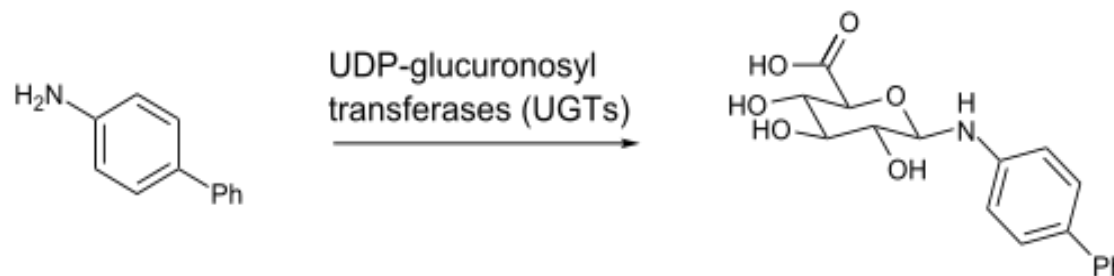
Reduction



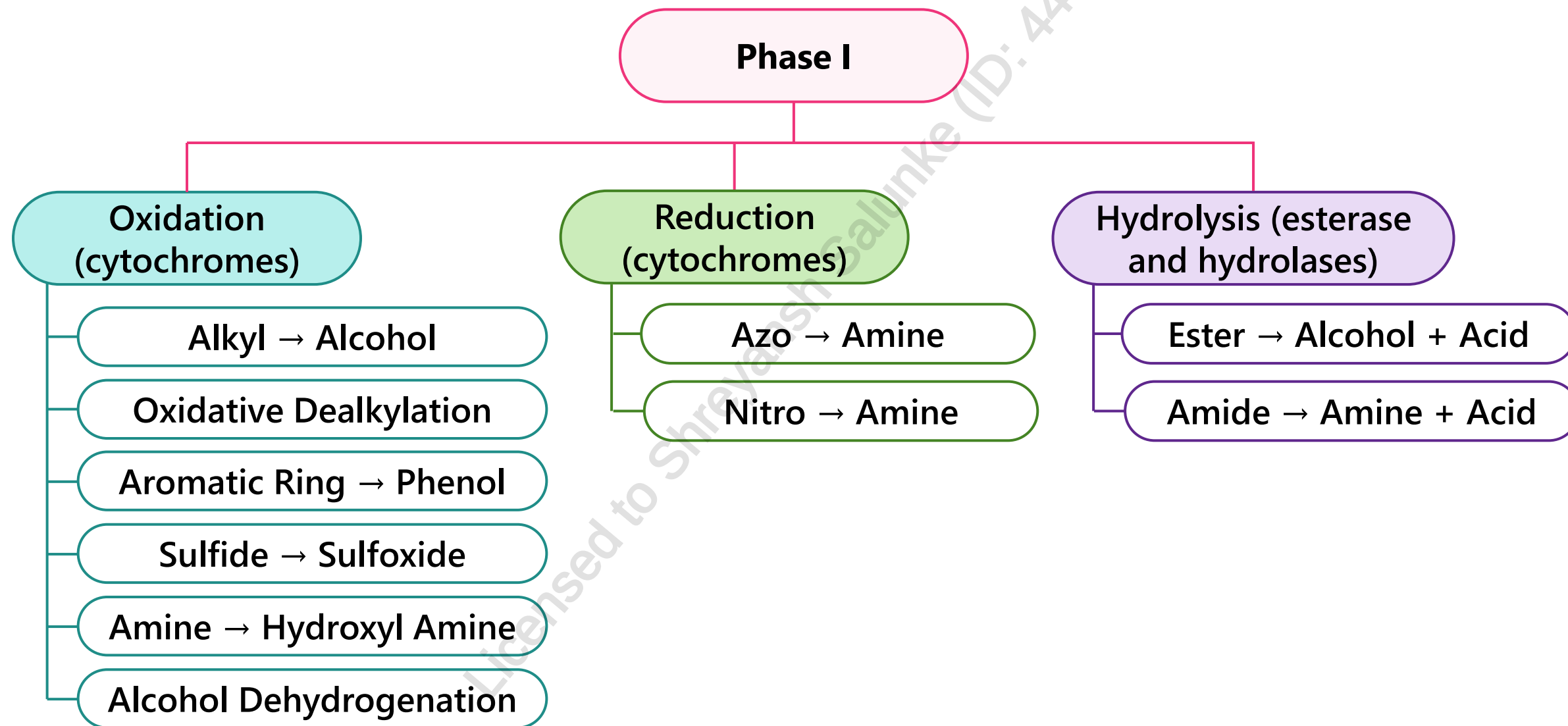
Hydrolysis



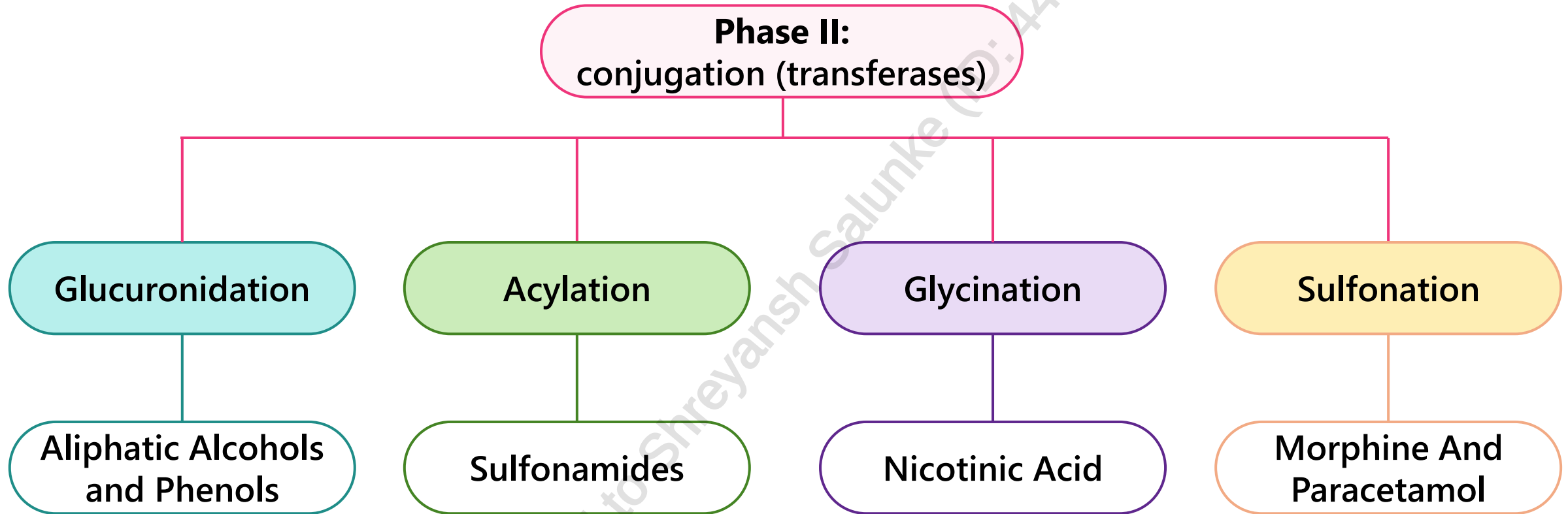
Conjugation



BIOTRANSFORMATION



BIOTRANSFORMATION



EXCRETORY CLEARANCE: AN OVERVIEW

- Biliary excretion: active transport of drugs into bile
- Often amphipathic drugs with large mass ($>300\text{g/mol}$)
- May be reabsorbed in GI tract or excreted via feces
- Drugs can also be excreted in urine, tears, saliva, sweat and milk



Bile



Sweat



Tears



Saliva

PHARMACOKINETIC INTERACTIONS

- Metabolism is the most common reason for drug interactions, mediated by phase I and II enzymes, and non-enzymatic processes
- Phase I isoenzyme system:
 - Cytochrome 450 (CYP450): hepatic and intestinally based, a family of oxidation enzymes that will prepare xenobiotics for further conjugation or direct elimination
 - CYP1A2, 3A4, 2C9/2C19, 2D6 and 2E1 are responsible for the majority of metabolism
 - MOST enzymes can be induced or inhibited
 - Substrates vary and may be inactivated or activated via CYP450 (e.g. clopidogrel)

CONSIDERATIONS WHEN INITIATING THERAPY

Acid-reducing agents

- ↓interactions: famotidine, ranitidine, lansoprazole, pantoprazole, rabeprazole
- ↑interactions: esomeprazole and omeprazole (2C9/2C19 inhibition), cimetidine (2C19/2D6 inhibition)
- Antacids (Ca^{2+} , Mg^{2+} , Al^{3+}) may bind to some drugs

Antibiotics: macrolides

- ↓interactions: azithromycin
- ↑interactions: clarithromycin and erythromycin (inhibit 3A4)

Antidepressants

- ↓interactions: citalopram, escitalopram, venlafaxine
- ↑interactions: bupropion, duloxetine, and paroxetine (inhibit 2D6); fluoxetine (inhibits 2C19, 2D6), fluvoxamine (1A2, 2C19), St John's Wort (CYP inducer)

Antiepileptics

- ↓interactions: gabapentin, lamotrigine, levetiracetam
- ↑interactions: carbamazepine, phenytoin, phenobarbital (CYP inducers)

CONSIDERATIONS WHEN INITIATING THERAPY

Contraceptives

- ↓interactions: copper IUD (minimal/no interactions)
- ↑interactions: combined oral contraceptives with CYP inducing drugs have a greater risk of contraceptive failure compared to levonorgestrel IUD and depo-medroxyprogesterone (IM)

Dyslipidemia

- ↓interactions: rosuvastatin, pravastatin
- ↑interactions: atorvastatin, lovastatin, simvastatin are CYP3A4 substrates; fluvastatin inhibit 2C9; gemfibrozil inhibits 1A2, 2C19, 2C8, 2C9 and may increase the serum concentrations of all statins (and increase the risk of rhabdomyolysis)

HIV agents

- ↓interactions: NRTIs (e.g. lamivudine, emtricitabine), tenofovir, raltegravir
- ↑interactions: NNRTIs (e.g. efavirenz, nevirapine), protease inhibitors (e.g. ritonavir, atazanavir), maraviroc

Opioids

- Methadone: many interactions (inhibits 2D6, 3A4 substrate), QT prolongation
- Fentanyl patch: caution with 3A4 inhibitors (reports of drug accumulation)

EXCRETION: MECHANISMS

Excretion most commonly results from inhibition of excretion by competitive or non-competitive means

**Organic Anion
Transporter Polypeptide
(OATP)**
e.g. cyclosporine, statins

PGP efflux inhibition

**Competitive renal
tubular excretion**
e.g. NSAIDs and
methotrexate, probenecid

**Co-transport of renal
excretion**
e.g. lithium and sodium

Biliary excretion

Renal ion trapping

MANAGEMENT OF INTERACTIONS

WITH INCREASED RISK OF TOXICITIES

Concern?	Management?
QT prolongation: • Torsades de pointes (ventricular tachycardia) may occur when using multiple drugs that prolong QTc	• Patients with multiple risk factors for prolonged QTC should have a baseline ECG. • Exercise caution if QTc ≥ 450 msec and avoid drugs if QTC ≥ 500 or >60 msec over baseline
Serotonin (5-HT) syndrome: • Symptoms: agitation, excitement, delirium, increased heart rate, myoclonus, tremor, ataxia, weakness, fever/chills, diarrhea, hyperreflexia (onset is within hours of adding/changing drug)	• Discontinue offending drug and manage symptoms. • Avoid combination therapy of MAOI and another serotonergic drug (e.g. SSRI)
Additive anticholinergic effects: • Decreased cognition, especially in elderly. • Examples drugs: antihistamines, benztropine, cyclobenzaprine, TCAs, paroxetine, prochlorperazine, some antipsychotics (e.g. clozapine, olanzapine)	• Discontinue offending medication, monitor, manage symptoms
Hyperkalemia: • Symptoms: muscle fatigue, nausea, weakness, paralysis, arrhythmias • Examples drugs: ACEI, ARB, cotrimoxazole, spironolactone, eplerenone, amiloride, triamterene, cyclosporine, tacrolimus, heparins, drospirenone, K⁺ supplements	• Monitor for weakness, paralysis, arrhythmias, nausea, muscle fatigue. May require hospitalization

MANAGEMENT OF INTERACTIONS

WITH INCREASED RISK OF TOXICITIES

Concern?	Management?
Additive Hypoglycemia Risk: Offending drugs: sulfonylureas, meglitinides, insulin, DPP-4 inhibitors, GLP-1 agonists, SGLT-2 inhibitors, quinine, cotrimoxazole	- Prevent hypoglycemia by educating patients on monitoring, and how to recognize and treat symptoms. - May require change in drug therapy
Increased Risk for Bleeding: Offending drugs: warfarin, anticoagulants, antiplatelets, NSAIDs, SSRIs, steroids, K⁺ supplements (GI bleeding), some natural health products	Assess if required on a regular basis and consider alternative therapy if possible. Consider gastroprotection (PPI) if high risk of GI bleeding
Lithium + NSAID/diuretic/ACEI/ARB → ↑ serum lithium levels by ↓ renal elimination by 25-50%	Monitor lithium serum levels (e.g. 3-5 days after initiating interacting treatment) and monitor for lithium toxicity (e.g. nausea, tremors, ataxia, drowsiness)
Azathioprine + Allopurinol → toxic levels of azathioprine and possible subsequent myelosuppression	If the combination is required, only use 25% of the usual azathioprine dose and monitor WBCs

MANAGEMENT OF INTERACTIONS

WITH THE POTENTIAL FOR LOSS OF EFFICACY

Concern?	Management?
Antacids, multivitamins, or other products containing Ca^{2+}, Fe^{2+}, Al^{3+}, Mg^{2+} + fluoroquinolones/tetracyclines/levothyroxine/bisphosphonates/dolutegravir → these minerals bind to and ↓ absorption of many drugs	Separate minerals and interacting medications by at least 2 hours
Acetylcholinesterase inhibitors (donepezil, galantamine, rivastigmine) + anticholinergic drugs (e.g. TCAs, antihistamines, antimuscarinics) → competing cholinergic and anticholinergic activity	Prioritize between cholinergic and anticholinergic activity (e.g. avoid using oxybutynin for treating urinary incontinence caused by donepezil)
Dopamine agonists (e.g. levodopa, pramipexole) + dopamine antagonists (e.g. antipsychotics, metoclopramide) → competing activity may ↓ therapeutic effectiveness	- In patients with Parkinson's disease avoid such combinations. - If an antipsychotic is necessary, consider using quetiapine or clozapine (lowest risk of interactions. Note, domperidone has a lower interaction risk than metoclopramide (lower blood-brain barrier penetration)

MANAGEMENT OF INTERACTIONS

WITH THE POTENTIAL FOR LOSS OF EFFICACY

Concern?	Management?
• Low-dose Aspirin® (81 mg)+NSAIDs → antiplatelet ability of Aspirin® is affected by concomitant NSAID use due to competition for platelet cyclooxygenase	• If appropriate, consider replacing the NSAID with acetaminophen. *Note, occasional single doses of NSAIDs (especially ibuprofen) administered at least 2 hours after Aspirin® have shown not to cause interactions. *Note, celecoxib does not cause this interaction however celecoxib on its own does have a high cardiovascular risk
• Rifampin + oral contraceptive → rifampin ↓ efficacy of oral contraceptives through CYP induction	• During rifampin treatment and for 7 days after, use second form of contraception

CYP1A2 INTERACTIONS

Inducers

- Carbamazepine
- Phenobarbital
- Rifampin
- Ritonavir
- Tobacco smoke (including cigarette smoke)
- Cannabis smoke

Inhibitors

- Amiodarone
- Cimetidine
- Ciprofloxacin
- Fluconazole
- Fluvoxamine
- Gemfibrozil
- Norfloxacin
- Ofloxacin

Substrates

- | | |
|-------------------|----------------|
| • Acetaminophen | • Fluvoxamine |
| • Amitriptyline | • Haloperidol |
| • Caffeine | • Imipramine |
| • Clomipramine | • Methadone |
| • Clopidogrel | • Olanzapine |
| • Clozapine | • Ondansetron |
| • Cyclobenzaprine | • Pomalidomide |
| • Desipramine | • Propranolol |
| • Diazepam | • Tizanidine |
| • Diphenhydramine | • Theophylline |
| • Duloxetine | • Verapamil |
| • Estradiol | • Warfarin |

CYP2C9 INTERACTIONS

Inducers

- Carbamazepine
- Dexamethasone
- Phenobarbital
- Phenytoin
- Rifampin
- Ritonavir
- St. John's Wort

Inhibitors

- Amiodarone
- Cotrimoxazole
- Delavirdine
- Efavirenz
- Etravirine
- Fluconazole
- Fluoxetine
- Fluvastatin
- Fluvoxamine
- Gemfibrozil
- Isoniazid
- Ketoconazole
- Metronidazole
- Omeprazole
- Paroxetine
- Quinine
- Sulfamethoxazole
- Ticagrelor
- Voriconazole

Substrates

- Amitriptyline
- Carvedilol
- Celecoxib
- Clomipramine
- Diazepam
- Diclofenac
- Diphenhydramine
- Doxepin
- Fluoxetine
- Glimepiride
- Glyburide
- Ibuprofen
- Imipramine
- Indomethacin
- Irbesartan
- Losartan
- Methadone
- Montelukast
- Naproxen
- Omeprazole
- Phenobarbital
- Phenytoin
- Rosiglitazone
- Sildenafil
- Tamoxifen
- Valsartan
- Vardenafil
- Warfarin

CYP2C19 INTERACTIONS

Inducers

- Carbamazepine
- Phenytoin
- Rifampin
- St. John's Wort

Inhibitors

- Cannabidiol
- Cimetidine
- Cinacalcet
- Delavirdine
- Efavirenz
- Esomeprazole
- Etravirine
- Fluconazole
- Fluoxetine
- Fluvoxamine
- Gemfibrozil
- Isoniazid
- Ketoconazole
- Moclobemide
- Modafinil
- Omeprazole
- Voriconazole

Substrates

- Amitriptyline
- Cannabidiol
- Citalopram
- Clobazam
- Clomipramine
- Clopidogrel
- Desipramine
- Diazepam
- Diphenhydramine
- Doxepin
- Escitalopram
- Indomethacin
- Methadone
- Phenobarbital
- Phenytoin
- PPIs
- Progesterone
- Propranolol
- Sertraline
- Warfarin

CYP2D6 INTERACTIONS

Inducers

- Not inducible

Inhibitors

- Amiodarone
- Bupropion
- Chloroquine
- Cimetidine
- Cinacalcet
- Clomipramine
- Delavirdine
- Diphenhydramine
- Dronedarone
- Duloxetine
- Fluoxetine
- Fluvoxamine
- Haloperidol
- Isoniazid
- Ketoconazole
- Methadone
- Mirabegron
- Paroxetine
- Quinine
- Quinidine
- Ritonavir
- Terbinafine
- Ticlopidine

Substrates

- Amitriptyline
- Amphetamine
- Aripiprazole
- Bisoprolol
- Carvedilol
- Chloroquine
- Clomipramine
- Clozapine
- Codeine
- Cyclobenzaprine
- Dextromethorphan
- Diphenhydramine
- Donepezil
- Doxepin
- Duloxetine
- Fesoterodine
- Flecainide
- Fluoxetine
- Fluvoxamine
- Haloperidol
- Hydrocodone
- Methamphetamine
- Metoprolol
- Mirtazapine
- Nortriptyline
- Olanzapine
- Ondansetron
- Oxycodone
- Paroxetine
- Propranolol
- Risperidone
- Ritonavir
- Tamoxifen
- Timolol
- Tramadol
- Trazodone
- Venlafaxine

CYP3A4 INTERACTIONS

Inducers

- Bosentan
- Carbamazepine
- Efavirenz
- Etravirine
- Modafinil
- Phenobarbital
- Phenytoin
- Rifampin
- St. John's Wort

Inhibitors

- Amiodarone
- Atazanavir
- Cannabidiol
- Clarithromycin
- Cobicistat
- Diltiazem
- Dronedarone
- Erythromycin
- Fluconazole
- Fluvoxamine
- Grapefruit
- Imatinib
- Itraconazole
- Ketoconazole
- Mifepristone
- Posaconazole
- Ritonavir
- Verapamil
- Voriconazole

Substrates

- Alfuzosin
- Alprazolam
- Amitriptyline
- Amiodarone
- Amlodipine
- Apixaban
- Aripiprazole
- Atorvastatin
- Budesonide
- Cannabidiol
- Carbamazepine
- Chloroquine
- Citalopram
- Clarithromycin
- Clonazepam
- Colchicine
- Cyclosporine
- Dexamethasone
- Diltiazem
- Domperidone
- Efavirenz
- Ergot derivatives
- Erythromycin
- Felodipine
- Fentanyl
- Fluticasone
- Haloperidol
- Hydrocodone
- Lovastatin
- Midazolam
- Mifepristone
- Nevirapine
- Nifedipine
- Oral contraceptives
- Oxycodone
- Paroxetine
- Quetiapine
- Rifabutin
- Risperidone
- Ritonavir
- Rivaroxaban
- Saxagliptin
- Sertraline
- Sildenafil
- Simvastatin
- Tacrolimus
- Tadalafil
- Tamoxifen
- Tamsulosin
- Ticagrelor
- Tolterodine
- Trazodone
- Triazolam
- Velpatasvir
- Verapamil
- Warfarin

CYP2E1 INTERACTIONS

Inducers

- Disulfiram
- Alcohol

Inhibitors

- Ethanol
- Isoniazid
- Fomepizole

Substrates

- Acetaminophen
- Alcohol

P-GP INTERACTIONS

P-GLYCOPROTEIN

P-GP is an efflux pump thus removes drugs from cells. The effects resulting from inhibiting or inducing P-GP depend on the location of the pumps in the body. In most situations, if P-GP is inhibited, drug levels are increased because the pump is highly prevalent along the intestinal tract.

Inducers

- Carbamazepine
- Dexamethasone
- Phenobarbital
- Phenytoin
- Primidone
- Rifampin
- St. John's Wort

Inhibitors

- | | | |
|------------------|----------------|---------------|
| • Amiodarone | • Indinavir | • Ritonavir |
| • Carvedilol | • Itraconazole | • Saquinavir |
| • Clarithromycin | • Ketoconazole | • Simeprevir |
| • Cobicistat | • Ledipasvir | • Tacrolimus |
| • Cyclosporine | • Mefloquine | • Tamoxifen |
| • Daclatasvir | • Mifepristone | • Telaprevir |
| • Diltiazem | • Nelfinavir | • Ticagrelor |
| • Dronedarone | • Paritaprevir | • Velpatasvir |
| • Duloxetine | • Posaconazole | • Verapamil |
| • Erythromycin | • Propafenone | |
| • Grapefruit | • Quinidine | |

Substrates

- | | |
|-----------------|----------------|
| • Amiodarone | • Nelfinavir |
| • Boceprevir | • Posaconazole |
| • Citalopram | • Ritonavir |
| • Cyclosporine | • Saquinavir |
| • Dabigatran | • Sertraline |
| • Dexamethasone | • Sofosbuvir |
| • Digoxin | • Telaprevir |
| • Diltiazem | • Velpatasvir |
| • Erythromycin | • Verapamil |
| • Indinavir | |
| • Lovastatin | |

OATP INTERACTIONS

ORGANIC ANION-TRANSPORTING POLYPEPTIDE

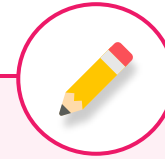
OATPs pump drugs into cells and are known as influx pumps. The effects resulting from inhibiting OATP depend on the location of the pumps: if located at the kidney inhibition causes increased drug levels, if located at the liver inhibition causes increased drug levels, if located in the GI tract inhibition causes decreased drug levels.

Inhibitors

- Clarithromycin
- Cyclosporine
- Erythromycin
- Gemfibrozil
- Grapefruit
- Ketoconazole
- Ritonavir
- Saquinavir
- Telaprevir
- Velpatasvir

Substrates

- ARBs
- Ciprofloxacin
- Erythromycin
- Montelukast
- Statins



Simvastatin is an OATP **hepatic** substrate. When administered with cyclosporine (OATP inhibitor), the simvastatin serum levels will be increased

CHANGE LOG

April 2025

- Slide 22: Added Tobacco “smoke” and cigarette smoke
- Slide 27: Removed cigarette smoke as it is primarily a CYP1A2 inducer (added to slide 22)

July 2025

- Content reviewed, minor formatting changes made

September 2025

- Slide 26: Added quetiapine, risperidone, rifabutin, rivaroxaban and ritonavir to substrates. Removed several drugs from list of inhibitors since no longer available on the Canadian market

October 2025

- Content reviewed, minor formatting changes made
- Slide 3: Added GI to legend
- Slide 9: Edited example and diagram
- Slide 13: added urine to the pathways of drug excretion
- Slide 22: Added cannabis smoke
- Slides 24 and 26: Added cannabidiol to lists of inhibitors and substrates

February 2026

- Slide 18: Updated inequality signs

March 2026

- Slide 2: Updated NAPRA competencies
- Slide 18: Minor update – changed inequality signs to ≥ 450 msec and ≥ 500 msec
- Slide 26: Added mifepristone to list of CYP3A4 substrates